

TECHNOLOGY



Designing Microsoft Azure Infrastructure Solutions: AZ:305

Design a Solution for Non-Relational Data



A Day in the Life of an Azure Architect

You are working as an architect in an organization, and you have been asked to choose the right type of Azure storage for the given use cases:

- To create an application that will be used to store the e-documents as part of the digital transformation journey in your organization
- To ensure that the overall cost of data storage is minimized, and the documents are stored in such a way that, depending on the document age, they move to the right tier of the storage
- To take care of the storage access by keeping data security in mind and providing some authentication method to access the data
- To build a solution for storing vast volumes of unstructured data

To achieve all the above, along with some additional features, you will be learning a few concepts in this lesson that will help in finding a solution for the above scenario.



Learning Objectives

By the end of this lesson, you will be able to:

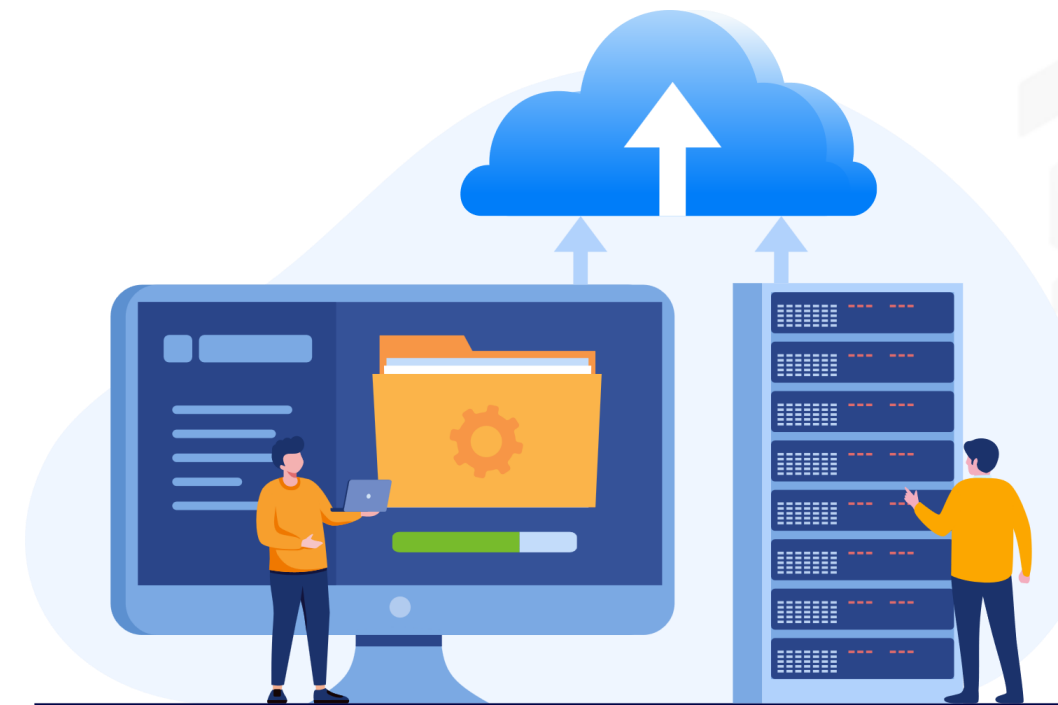
- Identify and explain the key features of Azure non-relational data to distinguish between different types and their optimal use cases
- Develop a design strategy for Azure Blob storage, applying best practices to optimize performance and efficiency
- Demonstrate proficiency in designing solutions for Azure files to showcase the ability to apply theoretical concepts to practical scenarios for effective file storage
- Formulate a comprehensive storage security plan for Azure, integrating appropriate measures to protect data integrity, confidentiality, and availability



Overview of Azure Non-Relational Data

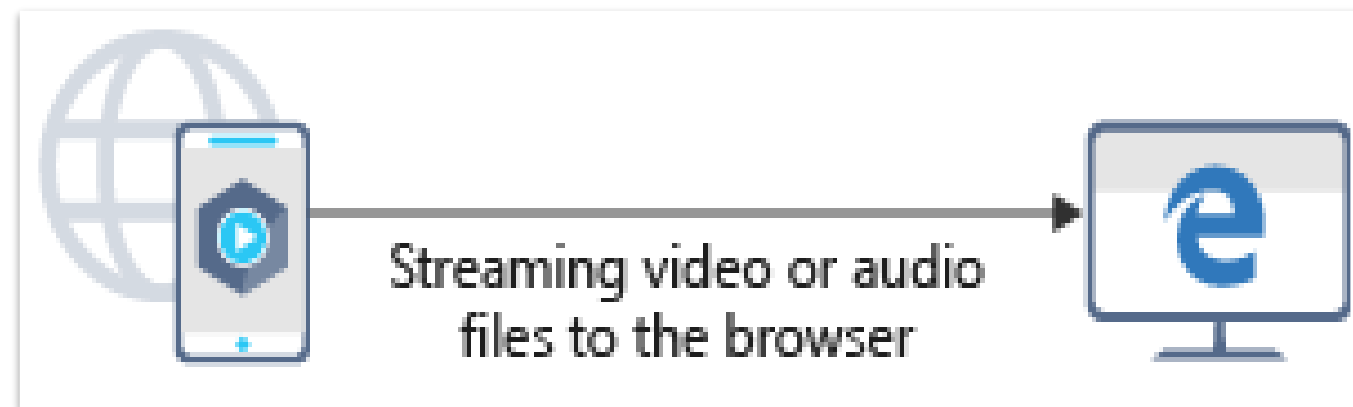
Azure Blob Storage

It aids in the creation of data lakes for analytics and provides storage for the development of strong cloud-native and mobile apps.



Azure Blob Storage

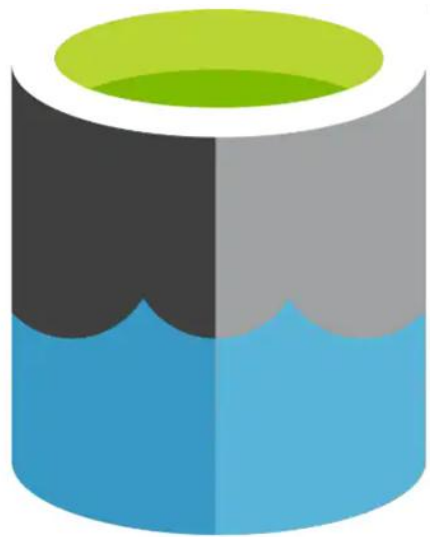
The features are as follows:



- Stores unstructured data storage
- Is highly scalable
- Is composed of thousands of simultaneous uploads
- Is accessible anywhere

Azure Data Lake Storage

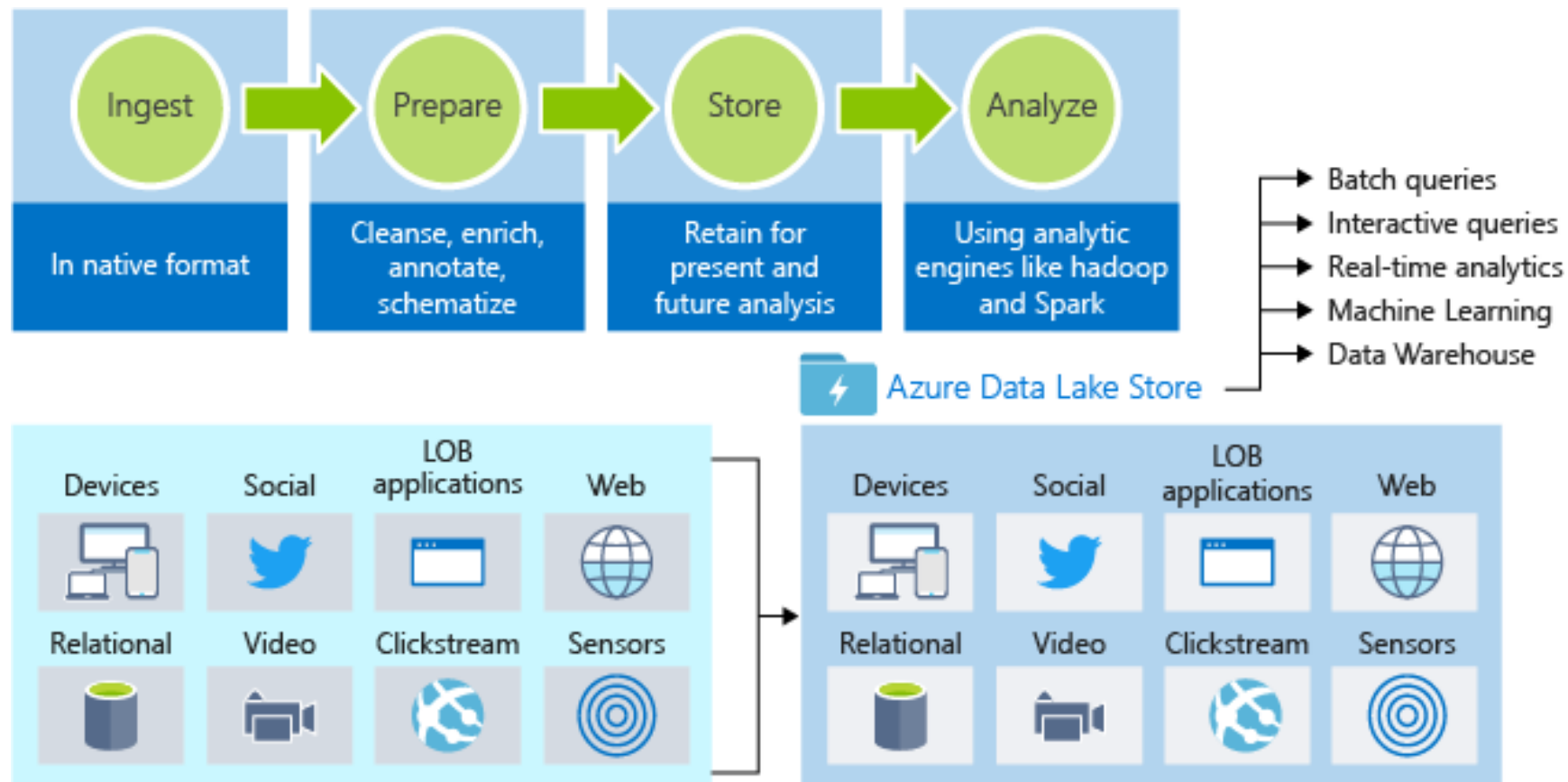
The Azure Data Lake platform is a cloud platform for big data analytics. It may store structured, semi-structured, or unstructured data indefinitely.



Azure Data Lake can store any data format, irrespective of the size of the company.

Azure Data Lake Storage

Azure data lake storage features are:



- Structured and unstructured data
- Hyperscale repository for big data analytics workloads
- No limits on account sizes, file sizes, or the amount of data

Candidate Services

Category	Azure Data Lake Storage Gen1	Azure Blob Storage
Purpose	Optimized storage for big data analytics workloads	General purpose object store for a wide variety of storage scenarios
Use cases	Batch, interactive, streaming analytics, and machine learning data such as log files, IoT data, click streams, and large datasets	Any type of text or binary data, such as application backend, backup data, media storage for streaming, and general-purpose data
Structure	Hierarchical file system	Object store with flat namespace
Authentication	Based on Microsoft Entra identities	Based on shared secrets: account access keys and shared access signature keys

Candidate Services

Category	Azure Data Lake Storage Gen1	Azure Blob Storage
Authentication Protocol	OpenID Connect. Calls must contain a valid JWT (JSON Web Token) issued by Azure Active Directory.	Hash-based Message Authentication Code (HMAC). Calls must contain a Base64-encoded SHA-256 hash over a part of the HTTP request.
Authorization	POSIX Access Control Lists (ACLs). ACLs based on Microsoft Entra identities can be set at the file and folder level.	For account-level authorization, use Account Access Keys.
		For account, container, or blob authorization, use Shared Access Signature Keys.
Auditing	Available	Available
Encryption data at rest	Transparent and server-side	Transparent and server-side or client-side
Developer SDKs	.NET, Java, Python, and Node.js	.NET, Java, Python, Node.js, C++, Ruby, PHP, Go, Android, and iOS

Candidate Services

Category	Azure Data Lake Storage Gen1	Azure Blob Storage
Analytics Workload Performance	Optimized performance for parallel analytics workloads with high throughput and IOPS	Optimized performance for parallel analytics workloads
Size limits	No limits on account sizes, file sizes, or number of files	For specific limits
Geo-redundancy	Locally redundant (multiple copies of data in one Azure region)	Locally redundant (LRS), zone redundant (ZRS), globally redundant (GRS), and read-access globally redundant (RA-GRS)

Azure Files

They are managed file shares in the cloud that are accessible via SMB. The files can be mounted concurrently by cloud or on-premises resources.

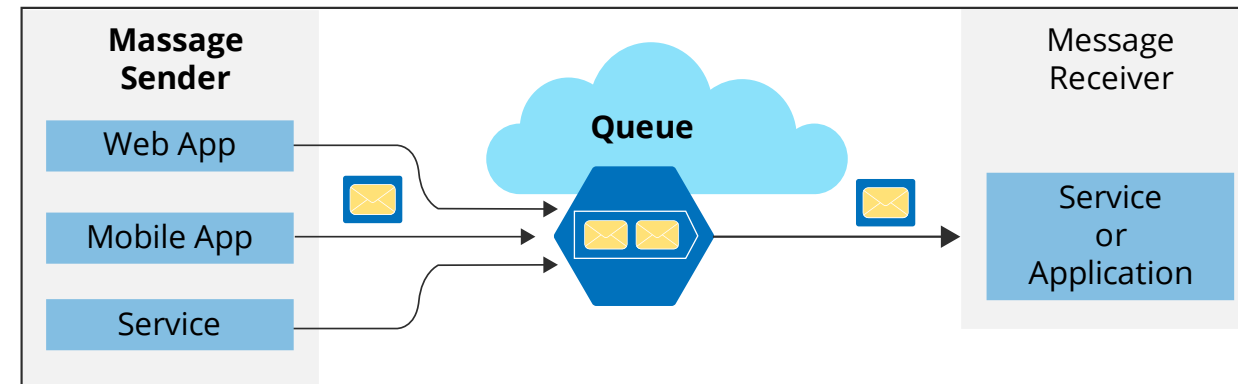


Common uses:

- Replace and supplement
- Lift and shift
- Azure File Sync
- Shared applications
- Diagnostic data
- Tools and utilities

Azure Queues

These are the features and uses of Azure queues:



Azure queues features:

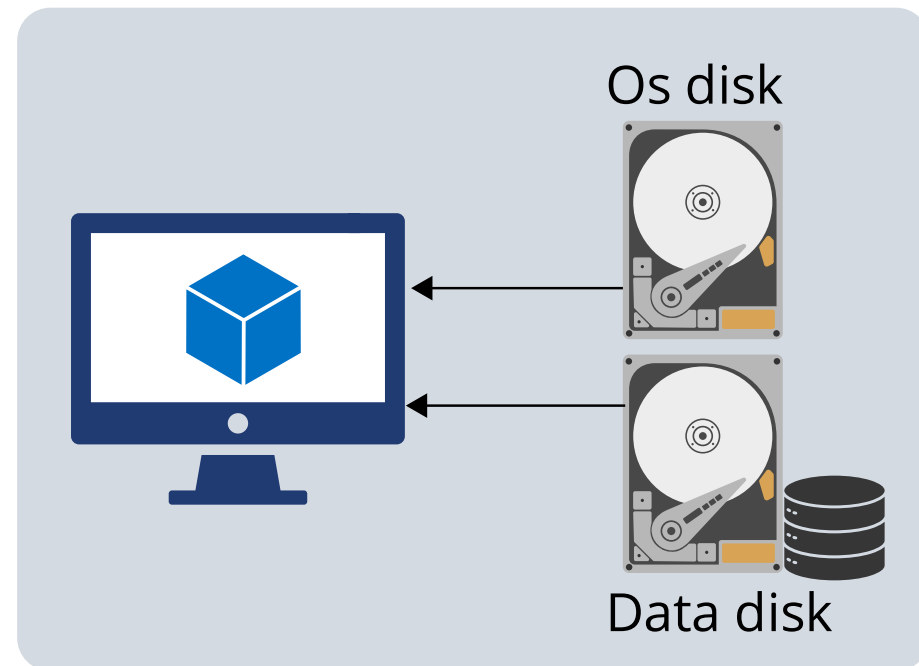
- Store large numbers of messages
- Can be accessed from anywhere in the world
- Provide asynchronous message queueing

Use queue storage to:

- Create a backlog of work
- Pass messages between services
- Distribute load
- Build resilience against component failures

Disk Storage

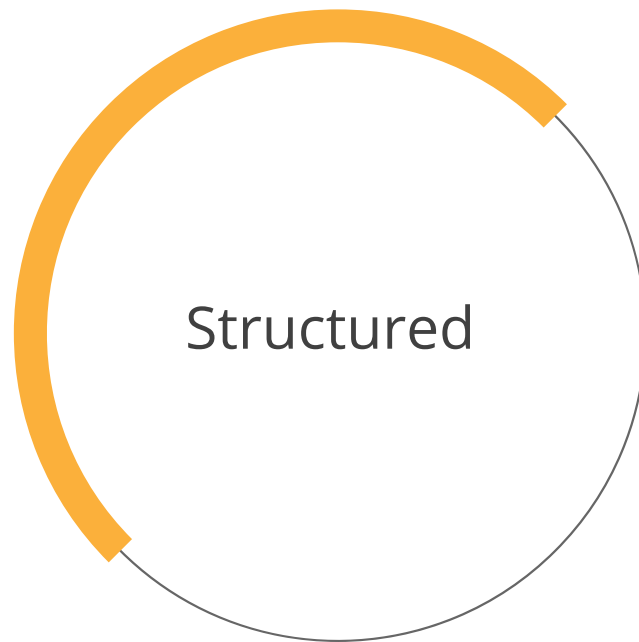
The disks can be managed or unmanaged. The following are the disk storage features:



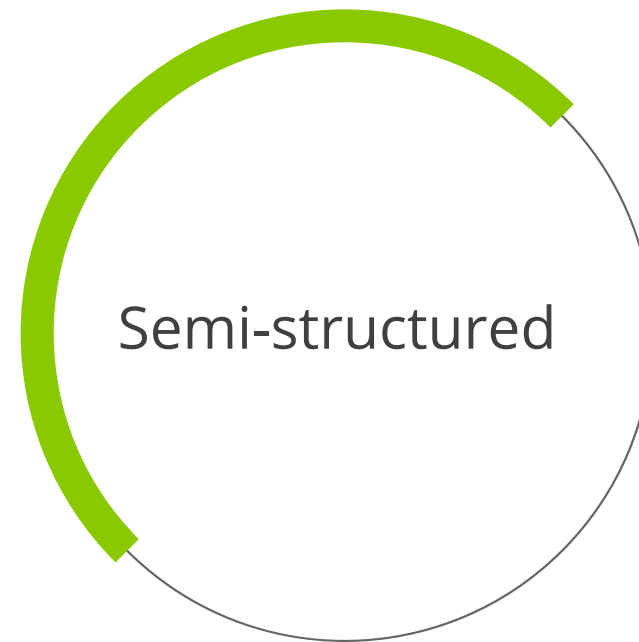
- It provides disks for virtual machines, applications, and other services.
- It has a persistent virtual hard disk storage.
- It has solid-state drives (SSDs) and hard disk drives (HDDs).
- It includes premium SSDs for mission-critical applications.

Classification of Data Storage

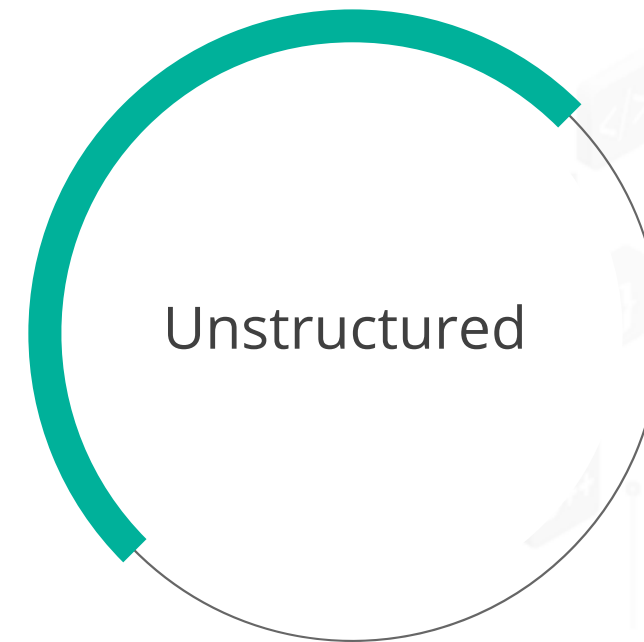
They are classified as shown below:



Characteristics:
Relational, tabular,
and follows a fixed
schema



Characteristics:
NoSQL, schema-less,
and supports semi-
structured data



Characteristics: No
specific data model
and supports any type
of data



Classification of Data Storage

Structured Storage:

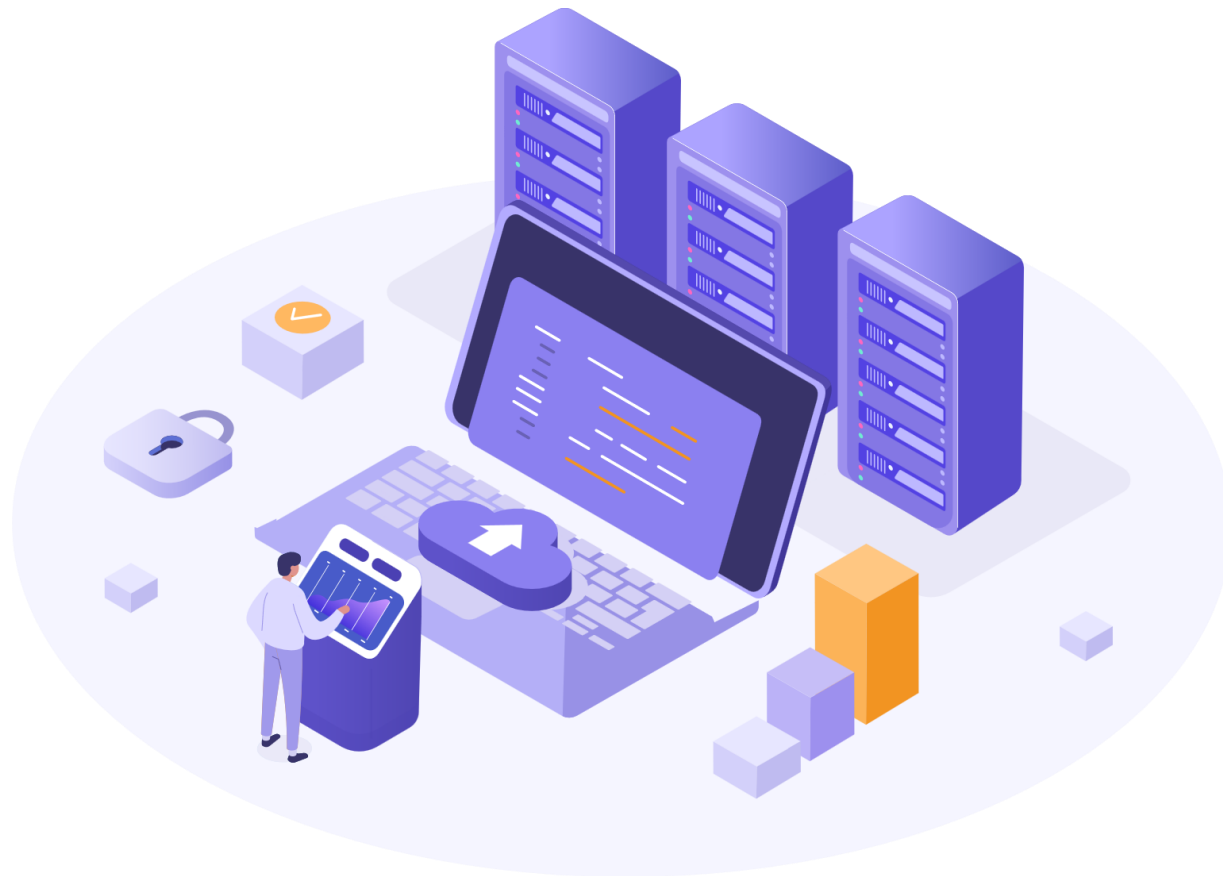
- Example: Azure SQL database
- Use case: Business applications and transactional data

Semi-Structured Storage:

- Example: Azure table storage
- Use case: IoT and JSON-based data

Unstructured Storage:

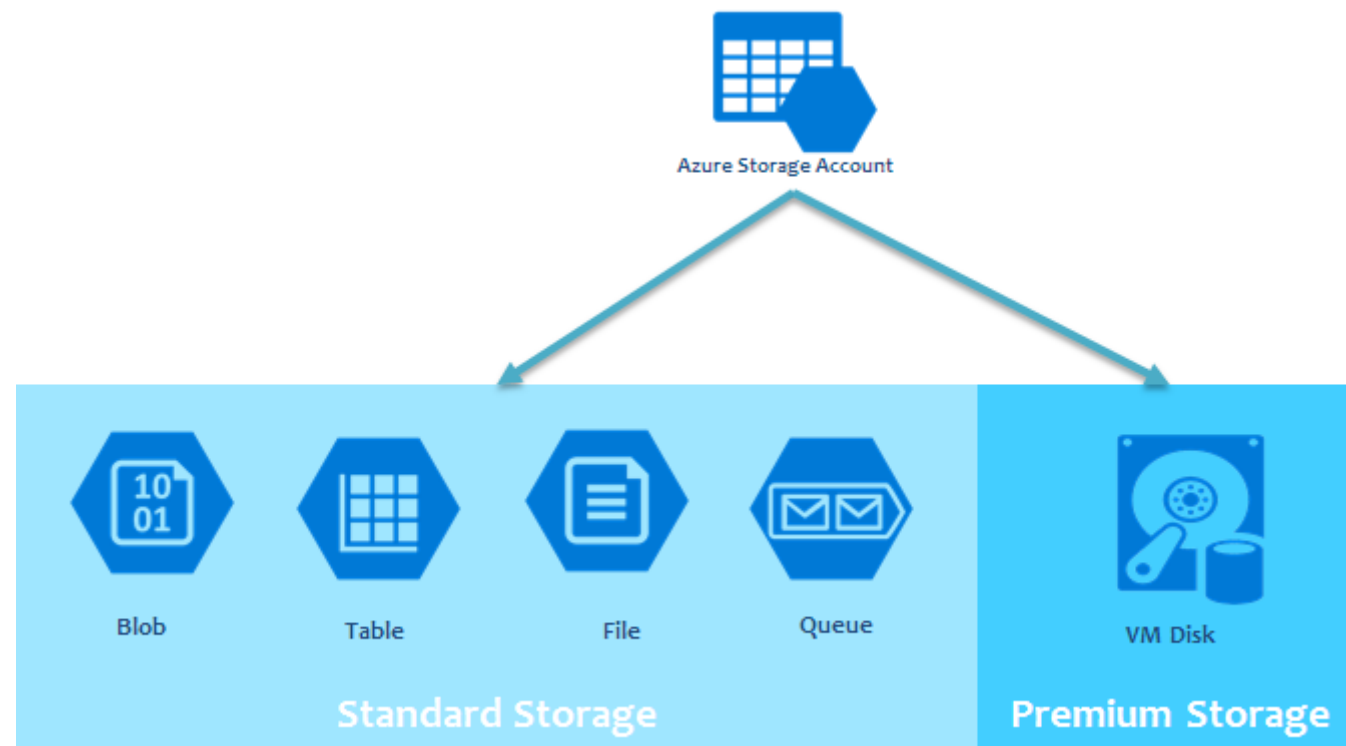
- Example: Azure blob storage
- Use case: Images, videos, documents, and backups



Storage Account

Azure Storage

It is a cloud storage platform designed for modern data storage scenarios.



Benefits:

- Durable and highly available
- Secure
- Scalable
- Automatic management
- Accessible

The service encrypts all data written to an Azure storage account.

Azure Storage Services

These are the Azure Storage services:

Azure Blobs

A text and binary data object store with huge scalability

Azure Files

A file share available for cloud or on-premises deployments

Azure Queues

A messaging store that allows application components to communicate reliably

Azure Tables

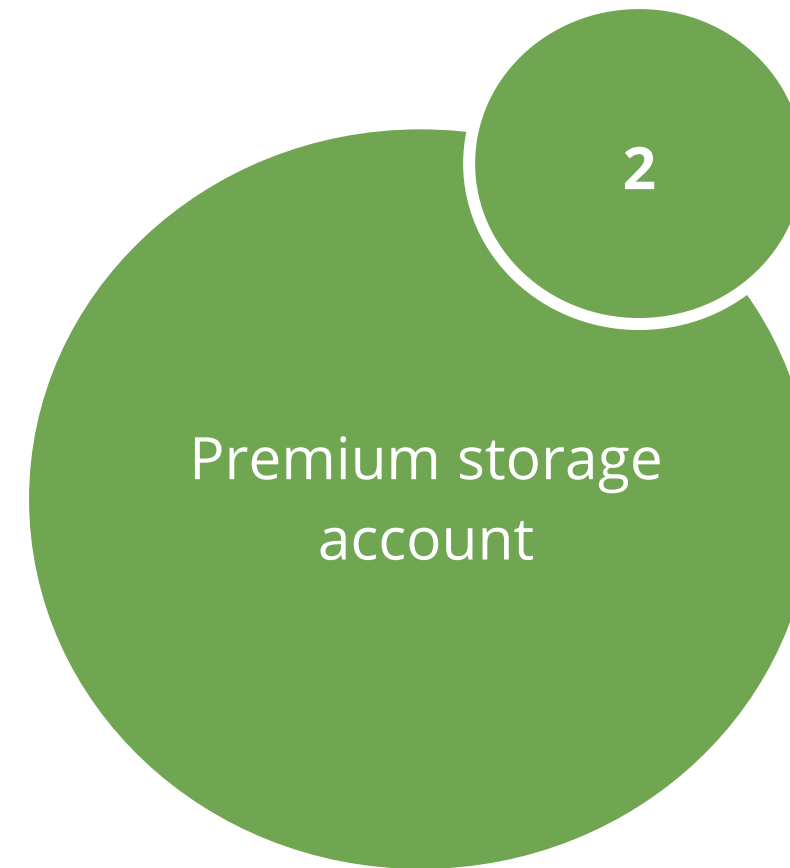
A schema-less NoSQL store for structured data storage

Azure Storage Services

The various Azure storage services are:

Storage Services	Description
Azure Blobs	Allow unstructured data to be stored and accessed as block blobs on a huge scale
Azure Files	Allow accessing fully managed cloud file shares from anywhere by using the industry standard Server Message Block (SMB) protocol
Azure Tables	Allow the user to store structured NoSQL data in the cloud and provide a schema-less key or attribute store
Azure Queues	Allow asynchronous message queuing between application components

Storage Account Types



Standard Storage Account

The standard storage performance tier can store tables, queues, files, blobs, and Azure virtual machine disks.



Most apps use standard storage, which is less expensive and slower.

Premium Storage Account

Premium disks aren't charged by transaction. It's more of a flat fee model.



Azure virtual machine disks are supported by the Premium storage performance tier only.



Storage Types

Standard
general-
purpose v2

Premium
block blobs

Premium file
shares

Premium
page blobs



Storage Types

Storage types	Supported storage services	Redundancy options
Standard general-purpose v2	Blob storage, queue storage, table storage, and Azure files	Locally redundant storage (LRS) / geo-redundant storage (GRS) / read-access geo-redundant storage (RA-GRS) Zone-redundant storage (ZRS) / geo-zone-redundant storage (GZRS) / read-access geo-zone-redundant storage (RA-GZRS)

Storage Types

Storage types	Supported storage services	Redundancy options
Premium block blobs	Blob storage	LRS ZRS
Premium file shares	Azure files	LRS ZRS
Premium page blobs	Page blobs only	LRS ZRS

Design for Azure Blob Storage

Blob Storage

It is an object storage solution for the cloud that is optimized for storing unstructured data.



Blobs

REST-based object storage for unstructured data

[Learn more](#)

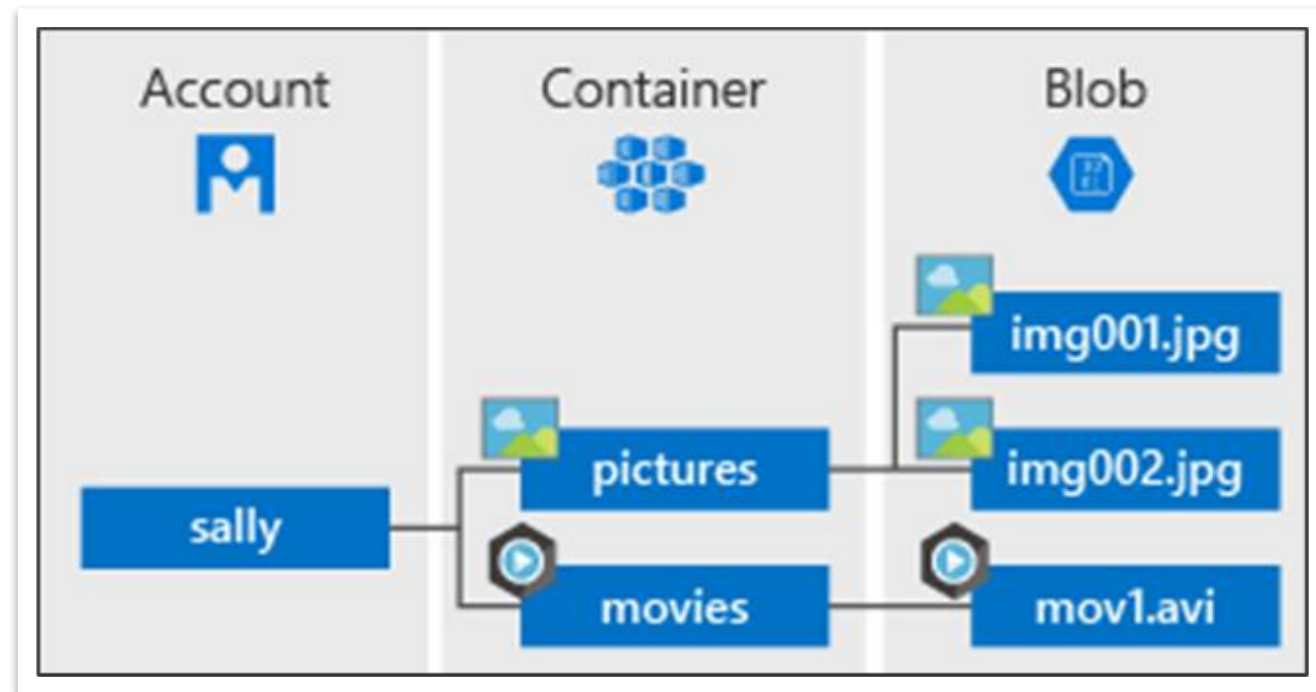
[Explore data using Azure AD preview](#)

It can store any type of text or binary data. It is also referred to as object storage.



Blob Storage

These are the common uses of blob storage:



- Serving images or documents directly to the browser
- Storing files for distributed access
- Streaming video and audio
- Storing data for backup and restore, disaster recovery, and archiving
- Storing data for analysis by an on-premises or Azure-hosted service

Blob Containers

The following are the features:



- A container organizes a set of blobs.
- A storage account can include an unlimited number of containers.
- A container can store an unlimited number of blobs.



Blob Performance Tiers

These are the blob performance tiers for storage:

Hot tier

Optimized for frequent access to objects in the storage account

Archive tier

Optimized for data that can tolerate several hours of retrieval latency and will remain in the archive tier for at least 180 days

Cool tier

Optimized for storing large amounts of data that is infrequently accessed and stored for at least 30 days

Cold tier

Optimized for storing data that is rarely accessed or modified but requires fast retrieval

Design for Blob Performance Tiers

The following points apply to the access tiers:



- At the account level, the hot and cool access tiers can be configured, but the archive access tier cannot.
- During or after upload, the user can organize the hot, cool, and archive levels at the blob level.
- The data in the cool access tier can withstand a minor reduction in availability.
- The archive storage stores data offsite and offers the most cost-effective storage options.

Support Tiering for Storage Accounts

These are the usage scenarios for different tiers:

Hot tier (inferred)

- It is used for storing data that is accessed or modified frequently.

Archive tier

- It is optimized for storing data that is rarely accessed.

Cool tier

- It is optimized for storing data that is infrequently accessed or modified.

Cold tier

- It is optimized for storing data that is rarely accessed or modified but still requires fast retrieval.

Uploading Blobs

The blob types that the user can upload are:

- **Block blobs (default):** Useful for storing text or binary files
- **Page blobs:** Efficient for frequent read/write operations
- **Append blobs:** Useful for logging scenarios

A user cannot change a blob type once it has been created.



Azure Files

Azure Files

It provides actively managed cloud file shares that can be accessed using the industry-standard Server Message Block (SMB) or Network File System (NFS) protocols.

It is useful to:

- Replace and supplement on-premise file servers
- Lift and shift applications
- Simplify cloud development
- Containerization



Azure Files vs. Azure Blobs

Feature	Description	When to use
Azure Files	SMB interface, client libraries, and a REST interface that allows access from anywhere to stored files	• Lift and shift an application to the cloud • Store shared data across multiple virtual machines • Store development and debugging tools that need to be accessed from many virtual machines
Azure Blobs	Client libraries and a REST interface that allows unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs	• Support streaming and random-access scenarios • Access application data from anywhere

Creating a File Share

► In Portal:

- When the Azure storage account deployment is complete, select Go to resource.
- Select File shares from the storage account pane.

► In PowerShell:

```
# Retrieve storage account and storage account key
$storageContext = New-AzStorageContext <storage-account-name>
<storage-account-key>

# Create the file share, in this case "logs"
$share = New-AzStorageShare logs -Context $storageContext
```



Mapping File Shares (Windows)

The methods to use an Azure file share with Windows are as follows:

- Mapping drive letter
- UNC path
- Account user
- Storage account key

Ensure port 445 is open.



Azure NetApp Files

It is a fully managed, high-performance file storage service.

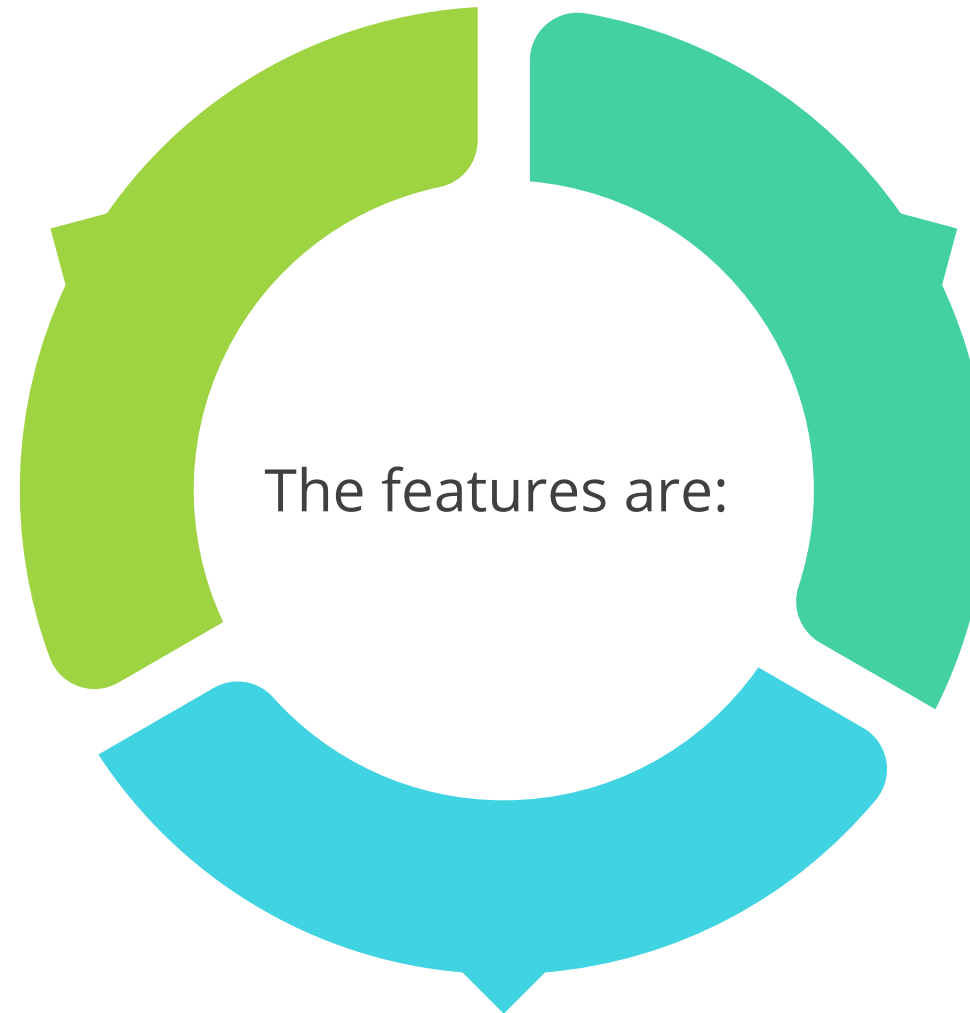


It is designed for enterprise-grade file workloads and supports complex analytics, databases, and applications.



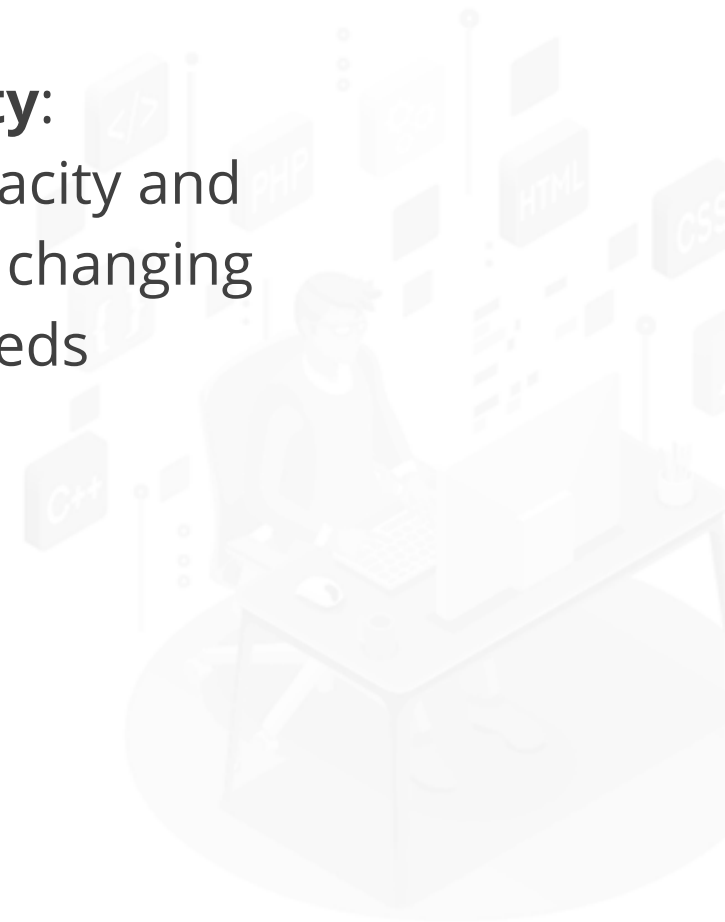
Azure NetApp Files

High Performance: Delivers sub-millisecond latency, ideal for high-performance applications



Scalability:
Easily scales capacity and performance for changing storage needs

Integration:
Has a seamless integration with Azure services and virtual networks



Azure NetApp Files

Use cases are as follows:

1

Is ideal for critical business applications requiring high performance and reliability

2

Supports large-scale database environments with consistent and low-latency performance

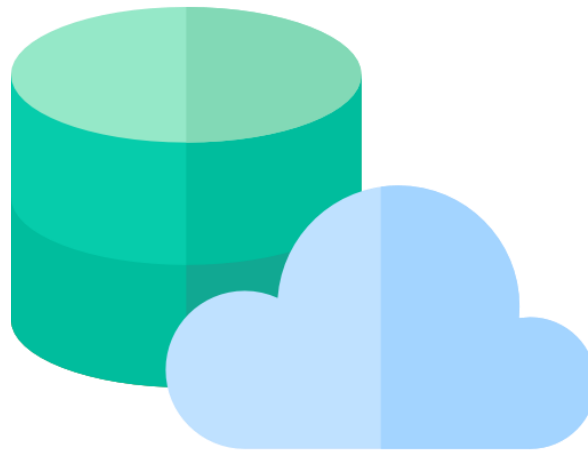
3

Enables fast and reliable data access for analytics and AI workloads

Design for Azure Disk Solutions

Data Disk

It is a managed disk that is attached to a virtual machine and used to store application data and other important information.

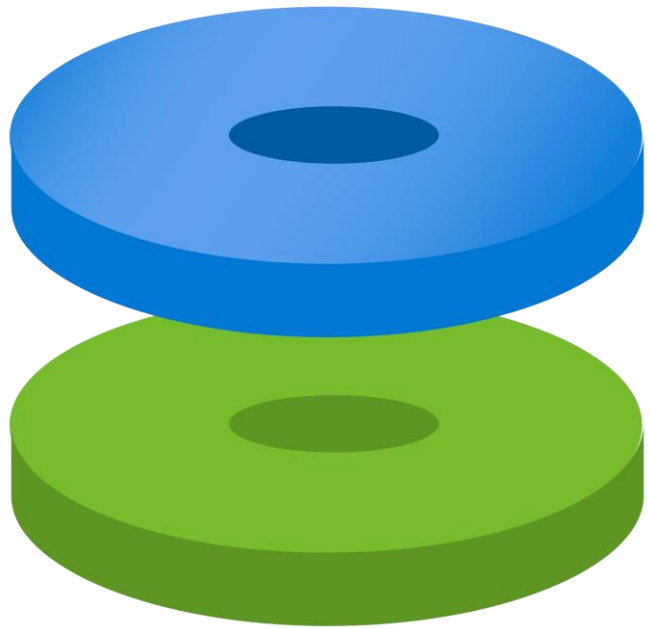


The maximum capacity of each data disk is 32 terabytes (TiB).

Data disks are registered as SCSI drives and are labeled with a letter that you choose.



Disk Types Available in Azure Managed Disks



Disk types are:

Ultra-disks

Premium solid-state drives (SSD)

Standard SSDs

Standard hard disk drives (HDD)

Data Disk

The following table provides a comparison of the five disk types:

	Ultra disk	Premium SSD v2	Premium SSD	Standard SSD	Standard HDD
Disk type	SSD	SSD	SSD	SSD	HDD
Max disk size	65,536 gibibyte (GiB)	65,536 GiB	32,767 GiB	32,767 GiB	32,767 GiB
Max throughput	4,000 MB/s	1,200 MB/s	900 MB/s	750 MB/s	500 MB/s
Max IOPS	160,000	80,000	20,000	6,000	2,000

Disk Caching

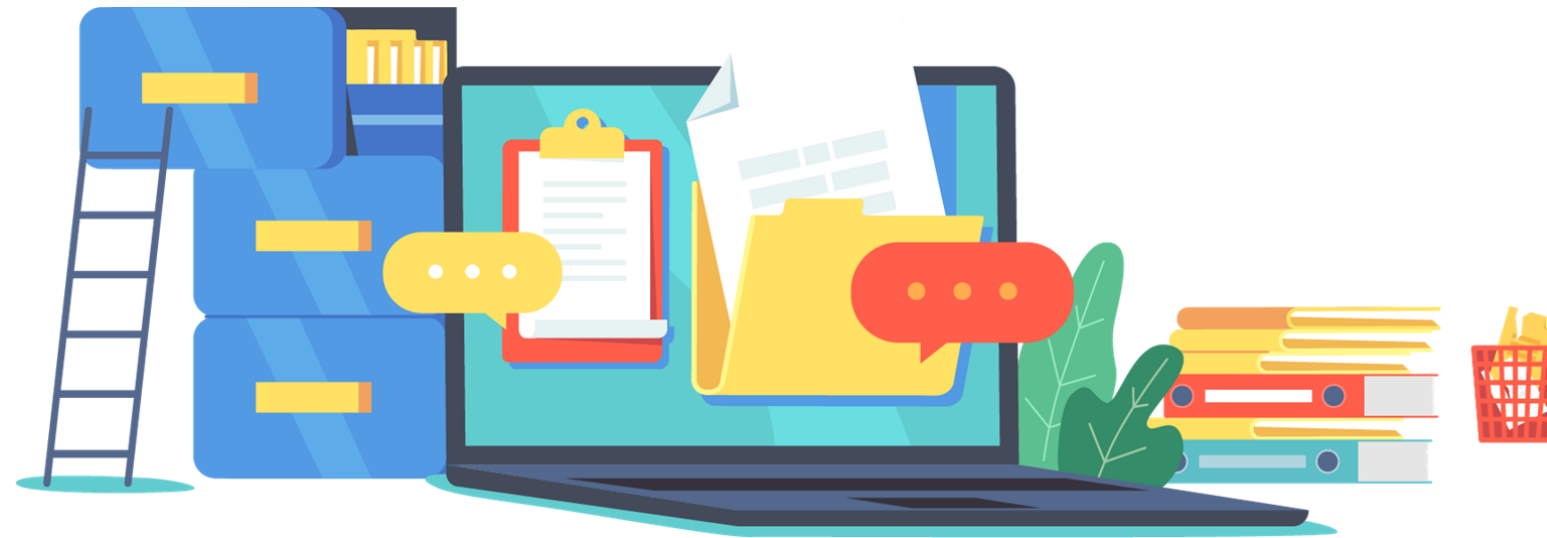
It is a method of reducing the time it takes to read or write data from or to a hard drive.



It is also a part of random-access memory (RAM) that has been set aside. It stores previously read data as well as data from nearby data regions that are expected to be accessed next.

Azure Disk Encryption

It assists the users in protecting and safeguarding their data to meet the organization's security and compliance obligations.



ADE encrypts the OS and data disks of Azure virtual machines (VMs) inside a user's VMs by utilizing the CPU of their VMs with Linux's DM-Crypt feature or Windows' BitLocker feature.

Storage Security

Storage Security

High-level security capabilities for Azure storage are:



- Storage encryption services
- Authentication with Microsoft Entra ID and RBAC
- Client-side encryption, HTTPS, and SMB 3.0 for data in transit
- Azure disk encryption
- Shared access signatures: delegated access

Storage Security



Authorization options:

Microsoft Entra ID

Shared key

Shared access signatures

Anonymous access to containers and blobs

Storage Account Keys

- Azure creates two keys, primary and secondary, for each storage account.
- Either of the keys provides full access to the account.
- Keys should be regenerated regularly or if they are compromised.



Shared Access Signature (SAS)

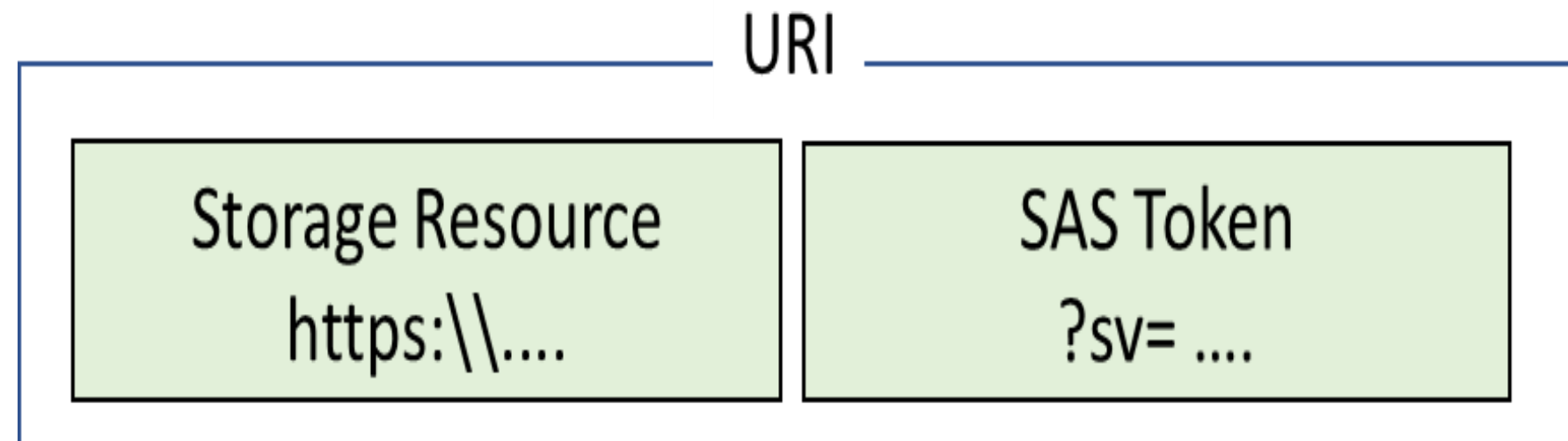
It provides delegated access to resources and grants access to clients without sharing storage account keys.



The account SAS delegates access to resources in one of the storage services, which are the Blob, Queue, Table, or File services.

URI and SAS Parameters

A SAS is a signed URI that points to one or more storage resources, and it consists of a storage resource URI and the SAS token.



They include parameters for resource URI, storage service version, services, resource types, start time, expiry time, resource, permissions, IP range, protocol, and signature.

Storage Service Encryption



- Protects data for security and compliance
- Encrypts and decrypts data
- Encrypts through 256-bit AES encryption
- Enabled for all new and existing storage accounts and cannot be disabled
- Transparent to users

Customer-Managed Keys



- Use the Azure Key Vault to manage the encryption keys.
- Distinguished personal encryption keys can be created and stored in a key vault.
- Azure Key Vault APIs are used to generate encryption keys.
- Custom keys give more flexibility and control.

Best Practices



Always use HTTPS to create or distribute a SAS

Reference stored access policies whenever possible

Use near-term expiration times on an ad-hoc SAS

Have clients automatically renew the SAS, if necessary

Be careful with SAS start time

Best Practices



Be specific about the resource to be accessed

Understand that your account will be billed for any usage

Validate data written using SAS

Don't assume SAS is always the correct choice

Use storage analytics to monitor your application

Managing Azure Storage



Duration: 10 Min.

Problem Statement:

You are given a project to manage Azure storage to secure and maintain data effectively within Microsoft's cloud storage services

Outcome:

You have successfully managed Azure storage, ensuring that the data is secure and properly maintained within Microsoft's cloud storage services.

Note: Refer to the demo document for the detailed steps
01_Managing_Azure_Storage

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a storage account
2. Manage blob storage
3. Create and configure Azure file share
4. Manage network access for Azure storage



Managing Storage Access Keys



Duration: 10 Min.

Problem Statement:

You are given a project to manage storage access keys to authorize access to data in your storage account via Shared Key authorization.

Outcome:

you have successfully managed the storage access keys and authorized access to the data in the storage account.

Note: Refer to the demo document for the detailed steps
02_Managing_Storage_Access_Keys

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. View and save storage account access keys
2. Install Azure Storage Explorer
3. Connect to Azure Storage



Key Takeaways

- Azure non-relational data provides a versatile and scalable solution for diverse data needs, offering flexibility, performance, and specialized features tailored to different use cases.
- Azure Blob storage aids in the creation of data lakes for analytics and provides storage for the development of strong cloud-native and mobile apps.
- Cloud or on-premises resources can concurrently mount managed file shares in the cloud that are accessible via SMB.
- Encrypting data in transit protects the data from outside observers and provides a mechanism for transmitting data while limiting the risk of exposure.





Knowledge Check

Knowledge Check

1

What is an Azure Storage Account?

- A. It contains one or more sets (containers) of your Azure Storage Services.
- B. It keeps key or value stores.
- C. It is an object storage solution provided by Azure.
- D. None of these



Knowledge Check

1

What is an Azure Storage Account?

- A. It contains one or more sets (containers) of your Azure Storage Services.
- B. It keeps key or value stores.
- C. It is an object storage solution provided by Azure.
- D. None of these



The correct answer is **A**

One or more sets (containers) of Azure Storage Services are stored in your Azure Storage Account (Azure Blobs, Azure Files, Azure Queues, and Azure Tables).

Knowledge Check

2

Which feature of Windows is used by Azure Disk Encryption?

- A. BitLocker
- B. Dm-crypt
- C. Both
- D. None



Knowledge
Check

2

Which feature of Windows is used by Azure Disk Encryption?

- A. BitLocker
- B. Dm-crypt
- C. Both
- D. None



The correct answer is **A**

Azure Disk Encryption uses BitLocker on Windows and the DM-Crypt feature of Linux.

**Knowledge
Check**

3

You must plan a method for decoupling application components and utilizing asynchronous data interchange and processing. What kind of storage service would you go with?

- A. Azure Blob storage
- B. Azure Queue storage
- C. Azure Table storage
- D. None of these



**Knowledge
Check**

3

You must plan a method for decoupling application components and utilizing asynchronous data interchange and processing. What kind of storage service would you go with?

- A. Azure Blob storage
- B. Azure Queue storage
- C. Azure Table storage
- D. None of these



The correct answer is **B**

Azure Queue Storage allows for asynchronous message queuing between application components.

Knowledge Check

4

How many storage tiers does Azure Blob Storage offer?

- A. 4
- B. 3
- C. 5
- D. 8



Knowledge Check

4

How many storage tiers does Azure Blob Storage offer?

- A. 4
- B. 3
- C. 5
- D. 8



The correct answer is **A**

Azure offers four storage tiers to store data in blob storage.

**Knowledge
Check**
5

For key management, which Microsoft Azure cloud security product can an organization use?

- A. Azure Guard Duty
- B. Azure Lockdown
- C. Azure Key Management
- D. Azure Key Vault



**Knowledge
Check**
5

For key management, which Microsoft Azure cloud security product can an organization use?

- A. Azure Guard Duty
- B. Azure Lockdown
- C. Azure Key Management
- D. Azure Key Vault



The correct answer is **D**

With Azure Key Vault, enterprises create and manage encryption keys. Admins can store and control access to items such as tokens, passwords, and API keys.

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Case Study

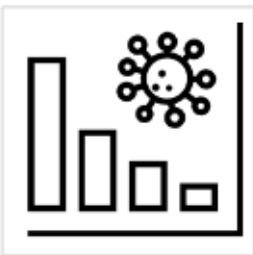
Design for Non-relational data Storage Solution

Scenario

Simplilearn wants to reduce storage costs by reducing duplicate content and, whenever applicable, migrating it to the cloud. They would like a solution that centralizes maintenance while still providing worldwide access for customers who browse media files and marketing literature. Additionally, they would like to address the storage of company data files.



Media files
• Product photos and feature videos • JPEG and MP4 are most common formats



Marketing literature
• Customer stories, sales flyers, sizing charts, and eco-friendly manufacturing information • PDF format is the most common



Corporate documents
• Internal documents – some sensitive • Mostly Office formats like Word and Excel

Design for Non-relational data Storage Solution

Scenario

- **Media files.** Media files include product photos and feature videos that are displayed on the company's public website, which is developed and maintained in house. When a customer browses to an item, the corresponding media files are displayed. The media files are in different formats, but JPEG and MP4 are the most common.
- **Marketing literature.** The marketing literature includes customer stories, sales flyers, sizing charts, and eco-friendly manufacturing information. Internal marketing users access the literature via a mapped drive on their Windows workstations. Customers access the literature directly from the company's public website.

Design for Non-relational data Storage Solution

Scenario

- **Corporate documents.** These are internal documents for departments such as human resources and finance. These documents are accessed and managed via an internally developed web application. Legal requires that various documents be retained for a specific period. Occasionally documents will need to be maintained longer when legal or HR issues are being investigated. Most corporate documents older than one year are only kept for compliance reasons and are seldom accessed.
- **File location:** All the files are stored locally in the main office data center. There are numerous file shares organized by department or product line. The data servers are struggling to provide files for the website. During peak hours website pages are slow to render.

Design for Non-relational data Storage Solution

Scenario

- **File access frequency:** Some products are more popular, and that data is accessed more frequently. However, some products, like ski gear, are only accessed during that season. Sales events generate a lot of interest in certain on sale items.

Design for Non-relational data Storage Solution

Requirements

1. Design a storage solution for **Simplilearn**.
 - What type of data is represented?
 - What factors will you consider in your design?
 - Will you use blob access tiers?
 - Will you use immutable storage?
 - How will the content be securely accessed?
2. Your solution should consider the media, marketing literature, and corporate documents. Your recommendations may be different depending on the data. Be prepared to discuss your decisions.

How are you incorporating the well-architected framework pillars to produce a high-quality, stable, and efficient cloud architecture?